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EmotionAI

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Abstract

The research aims to develop an EmotionAI system capable of accurately detecting human emotion through facial expressions in images, which is helpful in fields like healthcare, customer service, entertainment, and security. As technology continues to advance at a rapid pace, the significance of artificial intelligence has grown. Additionally, the potential of the application of AI grows as the virtual environment grows. These advancements allow for more interactions between humans, which foster an integration of AI, from healthcare to entertainment, and shape our daily lives. The hypothesis is that EmotionAI, using advanced facial recognition, can accurately identify and categorize positive and negative emotions, which will enhance different fields. The prototype of EmotionAI utilizes a model trained with large datasets of labeled facial images, which incorporates algorithms to analyze facial features such as eye shape, eyebrow position, and mouth corners. This system is built with OpenCV, Anaconda, and Jupyter and relies on image inputs to detect and classify emotions in real time. The result of the research is that EmotionAI demonstrates a high level of accuracy in identifying positive and negative emotions from facial expressions through the analysis of images. The system also categorizes emotions like happiness, sadness, and anger by interpreting key facial features and the potential for its use in healthcare, customer service, and security. This research concludes that EmotionAI is a field tool for accurately detecting and categorizing human emotions based on facial expressions and has significant potential in various industries. This research, EmotionAI, revolutionizes human-computer interactions, and future studies should focus on expanding the emotional detection capabilities and variations in facial expressions, and it will be extended for video inputs.

Introduction

The goal of the EmotionAI app is to design for human-computer interaction. It uses artificial intelligence to recognize, analyze, and respond to human emotions in real time. The app uses algorithms to detect emotional states from various inputs, such as facial expressions and features. By integrating multiple modes of emotion recognition, the system aims to provide users with detected emotion, whether to improve mental health, enhance communication in virtual environments, or provide emotional support. My Background Research is that as Artificial Intelligence grows, the integration of its applications is becoming impactful. After researching, I learned to use OpenCV, an open-source vision library, and enhanced facial recognition and emotion detection in images. I also learned how to use the combination of Anaconda and Jupyter, which facilitates data analysis and model training. This research is built upon advanced facial recognition algorithms that accurately detect and categorize emotions. The prototype of the EmotionAI app is designed to detect and categorize human emotions from facial expressions by using a combination of algorithms and a dataset of labeled emotions.

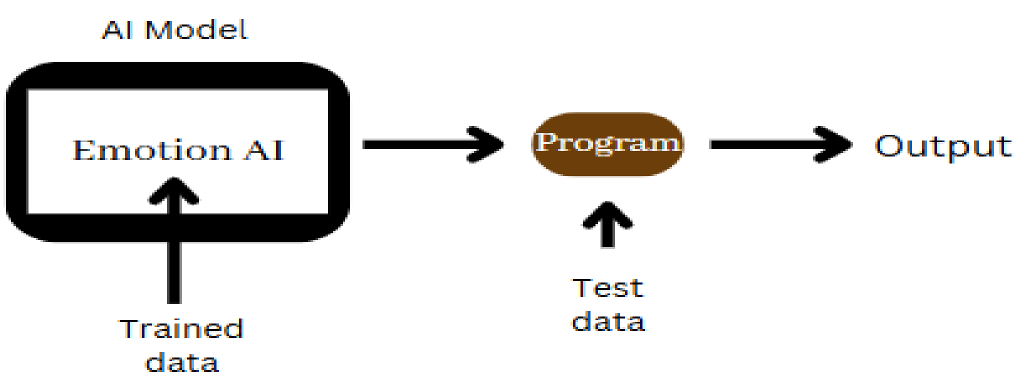

Figure 1: Architecture of Emotion AI Model

Methods and Materials

The materials used are:

- Laptop (I used a Lenovo ThinkPad W530)
- Python programming language
- Anaconda
- Jupyter Notebook
- Emotion Dataset – image data with labeled emotions
- Libraries: CV2, Matplotlib, DeepFace
- Internet connection

The materials are prepared as follows:

- Set up a computer by downloading Anaconda with Jupyter Notebook.
- Download the emotion detection model images and the labeled images data from Kaggle
- Open and launch Jupyter Notebook in Anaconda to write and execute Python code
- Prepare the environment by checking if the computer has enough storage space and power for training the model and running the emotion-detection system.

To begin the process, I first started by downloading Anaconda. I set up Anaconda on my computer and used Jupyter notebook as my main IDE for writing and executing the code. I began by importing libraries like OpenCV, Matplotlib, and DeepFace. Then, I downloaded pre-trained emotion detection model images from Kaggle, which contains images of faces labeled with their appropriate emotions. For the first image, I created an input to put the image name to search. Then, I trained it to read the image. After that, it will show the image. Then, the result is that it will read the face's emotions. Then, it will print the result. Finally, I trained it to print the dominant emotion. Then, this step is repeated for each image. The risks are that the app involves processing sensitive data, such as facial images. It is necessary to get the user's consent before capturing and processing their image for emotion detection. The second risk is algorithm bias. The emotion detection model could have biases based on the training data. To solve this risk, I ensured that the dataset was diverse and representative of different categories, such as genders and ages, to improve the predictions.

Presentation of Data and Results

The input data and the result of the model are given below.

```
img=input('Enter the image name \n')
print(img)
```

```
Enter the image name
 angry.jpg
angry.jpg
```

```
face1=plt.imread('angry.jpg')
plt.imshow(face1)
plt.show()
result1 = DeepFace.analyze(face1,actions = ['emotion'])
print(result1)
```


```
[{'emotion': {'angry': 92.35860796468162, 'disgust': 0.004255318201327948, 'fear': 6.466367814973853, 'happy': 0.1846410346831638, 'sad': 0.2310791243071
4078, 'surprise': 0.7425271486458601, 'neutral': 0.012512733483100417}, 'dominant_emotion': 'angry', 'region': {'x': 263, 'y': 133, 'w': 115, 'h': 115,
'left_eye': (342, 178), 'right_eye': (300, 176)}, 'face_confidence': 0.9}]
```

```
dominant_emotion = result1[0]['dominant_emotion']
print(dominant_emotion)
```

```
angry
```

Figure 2: Output for angry.jpg

This figure shows the input of the image “angry.jpg”, then it reads the image, and then displays the input image. Next, the system analyzes the facial features and prints the result. It displays a lot of emotions, like anger, disgust, fear, happiness, sadness, surprise, and neutral. Mainly, it displays the comment emotion, and then it prints the dominant emotion, which is “angry”. This trend is consistent with what would be expected, as anger is an easily identifiable expression characterized by furrowed brows, a tense mouth, and fists. However, the system also recorded some level of "Disgust" and "Fear," although at lower confidence levels, indicating some overlap in facial expressions associated with these emotions. Overall, the dominant emotion, "Angry," was successfully detected, with the system showing high accuracy in this particular test, while recognizing the nuances of other possible emotions as secondary or less intense.

```
img=input('Enter the image name \n')
print(img)
```

```
Enter the image name
 happy.jpg
happy.jpg
```

```
face3=plt.imread('happy.jpg')
plt.imshow(face3)
plt.show()
result3 = DeepFace.analyze(face3,actions = ['emotion'])
print(result3)
```

```
[{'emotion': {'angry': 1.7513657544226578e-09, 'disgust': 2.5303654349238737e-20, 'fear': 8.324852217438616e-10, 'happy': 99.98890161381979, 'sad': 6.277617841170627e-07, 'surprise': 0.008767054850173202, 'neutral': 0.0023310099145693415}, 'dominant_emotion': 'happy', 'region': {'x': 395, 'y': 122, 'w': 223, 'h': 223, 'left_eye': (547, 233), 'right_eye': (474, 198)}, 'face_confidence': 0.96}]
```

```
dominant_emotion = result3[0]['dominant_emotion']
print(dominant_emotion)
```

```
happy
```

Figure 3: Output for happy.jpg

This figure shows the input of the image “happy.jpg”, then it reads the image, and then displays the input image. Next, the system analyzes the facial features and prints the result. It displays a lot of emotions, like anger, disgust, fear, happiness, sadness, surprise, and neutral. Mainly, it displays the dominant emotion, and then it prints the dominant emotion, which is “happy” with the system reporting a high confidence score, as expected for a strong and easily recognizable facial expression, characterized by a wide smile and raised cheeks. A significant trend observed was that emotions like "Neutral" and "Sad" also had some level of detection, but at much lower confidence levels, which is typical for emotions with less intense facial features compared to happiness. Overall, the results showed a high degree of accuracy in identifying happiness, with the system confidently categorizing the dominant emotion as "Happy" and effectively discarding the other less prominent emotions.

Confusion matrix:

- Out of 50 test images, 49 give true positives, and 1 gives a false positive.

```
[[15  0  0  0  0  0]
 [ 0  5  0  0  0  0]
 [ 0  0  5  0  0  0]
 [ 0  0  0 14  0  1]
 [ 0  0  0  0  5  0]
 [ 0  0  0  0  0  5]]
```

The classification report shows that the accuracy of this model for the test images is 98%.

	precision	recall	f1-score	support
angry	1.00	1.00	1.00	15
disgust	1.00	1.00	1.00	5
fear	1.00	1.00	1.00	5
happy	1.00	0.93	0.97	15
sad	1.00	1.00	1.00	5
surprise	0.83	1.00	0.91	5
accuracy			0.98	50
macro avg	0.97	0.99	0.98	50
weighted avg	0.98	0.98	0.98	50

Interpretation of Results and Discussion

The goal of this project was to design and develop an emotion detection AI app that accurately analyzes facial expressions, identifying the user's emotional state from a set of 6 predefined emotions by importing an image. The system is expected to work with a pre-trained model to classify emotions based on facial features. Based on the test results, the EmotionAI app prototype functioned as anticipated, with some key successes and areas for improvement. The accuracy of emotion detection was generally strong, with the app correctly identifying emotions like happiness, sadness, and anger in most cases. However, there were occasional challenges with more subtle or mixed emotions, which revealed the need for improvement in the algorithm for better handling of unclear expressions. Overall, while the prototype met many expectations, further fine-tuning is required to enhance emotion recognition accuracy and optimize user interaction, especially in complex real-world environments. The results obtained from the EmotionAI prototype can be explained by a combination of algorithmic design and machine learning. The system's ability to detect basic emotions like happiness, sadness, and anger is based on established facial expression analysis models, which rely on identifying key facial features and comparing them to pre-trained datasets. However, more complex or subtle emotions, such as confusion or mixed feelings, may be challenging due to the limitations of current models in distinguishing between expressions. The results from the testing process demonstrate that the prototype largely addressed the engineering design statement by providing emotion detection through facial feature analysis. However, the accuracy and reliability of the prototype were affected by certain environmental factors, such as lighting and image quality, which were not fully anticipated in the original design. The system was able to detect basic emotions like "happy," "sad," and "angry" with reasonable precision, but improvements in model training and hardware would increase the overall accuracy of the system.

Conclusion

The main success would be how accurate the emotion recognition is. If the app is successfully identifying emotions, then it shows a solid foundation for the AI. Another success is that this app helps engage users by offering insights and helping them understand their emotions and any patterns or triggering factors, so it could foster better mental well-being. As I was developing the app, there were several areas of further investigation, like some questions and ideas. Some questions are: How do different demographics, like age, gender, and culture, affect the accuracy of emotion detection algorithms, and does it even affect in the first place? Can the app detect mixed or complex emotions? How can the app offer real-time emotional support, and Is effective feedback provided when it detects distress or negative emotions? To answer this question, I got an idea that has emerged in the app. The idea is to offer real-time feedback like providing guided meditation, positive affirmations, or calling emergency contacts or mental health resources if there are severe critical levels of distress detected. The main benefit is the potential application in mental health, and it could inspire scientists to explore further how AI could support emotional well-being and offer appropriate resources for mental health conditions. This research significantly contributes to the development and improvement of AI models used in emotion detection.

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